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QI RECEIVER 5V (BQ51013B) + 20mA CHARGER (MCP73832)

BATTERY + 3.3V LDO + VBAT MONITOR

nRF52832 CHIP-DOWN + 32MHz XTAL + RF FRONT-END + SWD

ANALOG FRONT-END: 2-OP-AMP POTENTIOSTAT (+0.512V)

DISCRETE POTENTIOSTAT – wireless–charged 3–electrode chronoamperometry

E_{cell} = WE – RE = 1.024V – 0.512V = +0.512V fixed. I_{WE} = (V(ADC_P)–V(ADC_N))/Rf

DESIGN NOTES / VERIFY BEFORE LAYOUT

- CHARGING: two-stage per TI guidance for small cells. BQ51013B = Qi receiver with regulated 5V (VSOUT). ILIM: RILIM = 250/IMAX = R1+R2 = 2.52k -> IMAX 100mA (HW limit 120mA). MCP73832–2 (4.2V) charges the LIR2032: ICHG = 1000/RPROG = 1000/49.9k = 20mA (0.5C), termination -7.5% = 1.5mA, open-drain STAT -> CHG_N (100k pullup to 3V3). This replaces the earlier BQ51050B direct-charge (TI: not recommended <200mA).
- RESONANT TANK: L1 11uH (TDK WR202010–18M8–ID, 20x20x1.0mm, DCR 350mR). C1 = 1/(L*(2pi*100kHz)^2) = 230n -> 220n COG (fs 102kHz). C2 -> fd -1MHz -> 2.2n COG. Retune both whenever the coil changes. Small coil = tighter TX alignment; fine at our 0.13W.
- CHIP-DOWN RF: copy Nordic's QFN48 reference layout verbatim (trace geometry included). FL1 (Johanson 2450FM07A0029) replaces the discrete LC match for nRF52 single-ended ANT. C35/R15/C34 = pi tuning network (shunt-series-shunt) REQUIRED by the 2450AT07A0100 datasheet for non-eval PCBs: start DNP/OR/DNP, tune with VNA on the eval board (eval board 40x20mm); keep battery can and Qi coil out from under this corner (shift E1 keypad: ~5x3mm copper-free zone on ALL layers at the board corner (eval board 40x20mm)).
- TS/CTRL: 10k NTC against the LIR2032 holder cuts wireless power if the cell heats (plain 10k = sense defeated, still a valid safe strap). EN1=EN2=GND: wireless enabled.
- AFE_PWR is driven by nRF GPIO P0.06 set to high-drive. Load = REF35+OPA2391 - 60uA. Power the AFE off between runs to avoid electrode polarization.
- WE net: guard ring driven from VREF, no soldermask opening at TIA input, clean flux.
- RF/CF options: 1M/470p (default, +2.1uA/-0.9uA FS), 5.1M/100p (427/181nA), 10M/47p (217/92nA). r_lohm + oversample are runtime settings in firmware; recalibrate after a swap.
- R10/C21 control-amp compensation are DNP; fit only if the CE-RE loop oscillates.
- LIR2032 has no protection circuit: firmware must System-OFF below 3.0V (VBAT_DIV, AIN2).
- SAADC: differential P=AIN1(ADC_P) N=AIN0(ADC_N), gain 1/2..4, oversample 64x default.
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